OSSP

LRU Page Replacement Algorithm

team 14

2510030179 CH Hymavathy
2510030404 Jyoshna sri
2510030316 Avusula Rajeshwari
2510030397 R Jashwitha

Problem Statement

To design and implement an LRU (Least Recently Used) Page Replacement system that efficiently manages memory by identifying and replacing the page that has been least recently accessed, thereby reducing page faults and improving memory utilization.

Introduction

- LRU stands for Least Recently Used.
- It is a page replacement algorithm.
- It is used when the main memory is full.
- LRU replaces the page that has not been used for the longest time.
- The main aim is to reduce page faults.

Objective

- To understand page replacement in Operating Systems.
- To implement the LRU algorithm.
- To identify page hits and page faults.
- To reduce the number of page faults.
- To improve memory management.

Logic / Working

- Take the reference string as input.
- Set the number of available frames.
- Check whether the requested page is already in memory.
- If it is present → Page Hit.
- If it is not present → Page Fault.
- If the frames are full, find the least recently used page.
- Replace that page with the new page.
- Continue until all pages are processed.

Advantages

- Simple concept.
- Uses recent page history.
- Can reduce page faults.
- Better than some basic replacement methods.

Disadvantages

- Requires tracking page usage.
- Needs additional memory/time.
- Implementation can be more complex than FIFO.

Conclusion

- LRU replaces the least recently used page.
- It helps manage limited memory.
- It identifies page hits and page faults.
- The project demonstrates how LRU works using sample inputs.

Thank You